

Solving Absolute Value Linear Inequalities

TEACHER KEY | Algebra II Honors | Unit 2 | Day 7 | TEK A2.6.F

Objective: I can justify why an absolute value inequality splits into an "and" or an "or" compound inequality, using the distance meaning of absolute value, and solve harder cases.

Vocabulary

Compound inequality — two inequalities joined by "and" or "or".

Part 1 — Why "And" and Why "Or"? Think Distance

Recall from Day 1: $|x|$ is the distance of x from 0. Use that to explain $|x| < k$.

$|x| < k$ means "x is within distance k of 0" - closer to 0 than k in EITHER direction, so x must be **BETWEEN** -k and k. That is why $<$ gives an **AND** statement.

Now explain $|x| > k$.

$|x| > k$ means "x is FARTHER from 0 than k" - either past k on the positive side, **OR** past -k on the negative side. That is why $>$ gives an **OR** statement.

Part 2 — Apply It

Example 1 ($<$, AND). $|2x + 3| < 5$

$$-5 < 2x + 3 < 5 \rightarrow -4 < x < 1$$

Example 2 ($\geq$, OR). $|2x - 3| + 4 \geq 9$

$$\text{Isolate: } |2x - 3| \geq 5 \rightarrow x \leq -1 \text{ or } x \geq 4$$

Part 3 — Harder Cases

Example 3. $|3 - 2x| \leq 7$ (the "inside" is not in the usual $x - c$ order)

$$-7 \leq 3 - 2x \leq 7 \rightarrow -10 \leq -2x \leq 4 \rightarrow \text{dividing by } -2 \text{ flips both} \rightarrow -2 \leq x \leq 5$$

Example 4. $5|x + 1| - 3 > 12$

$$\text{Isolate: } |x + 1| > 3 \rightarrow x < -4 \text{ or } x > 2$$

You Try. $2|x - 4| + 1 \leq 9$

$$|x - 4| \leq 4 \rightarrow 0 \leq x \leq 8$$

Practice

1. $|4 - x| > 6$ **$x < -2$ or $x > 10$**

2. $3|x - 2| < 12$ **$-2 < x < 6$**

3. Explain, using distance, why $|x + 2| \geq 0$ is true for EVERY real number x - no algebra needed.

distance from any point to a fixed point is never negative, so it is always ≥ 0

Exit Ticket

Solve $|3x + 2| \geq 7$. Then, without solving, explain using distance why $|x - 5| < -2$ has no solution at all.

$x \leq -3$ or $x \geq 5/3$. Distance can never be less than a negative number like -2, since distance is never negative